

PHISHING EMAIL DETECTION & AWARENESS REPORT

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FUTURE INTERS

Prepared by: Saad Ahmed

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1. Executive Summary

Phishing remains one of the most prevalent and successful forms of cyberattack, targeting individuals and organizations through deceptive emails that imitate trusted brands and services. These attacks exploit human psychology by creating a sense of urgency, fear, or curiosity to persuade recipients to reveal sensitive information, click malicious links, or contact fraudulent support services. Due to the increasing sophistication of phishing campaigns, user awareness has become a critical component of modern cybersecurity strategies.

This project was conducted as part of the **Future Interns Cyber Security Internship Program** with the objective of analyzing real-world phishing email samples and identifying the indicators commonly used by attackers. The project focuses on understanding how phishing emails are structured, recognizing suspicious characteristics, assessing the level of risk associated with each email, and providing practical recommendations to help users identify and avoid phishing attacks.

A total of **five phishing email samples** representing different attack scenarios—including credential theft, account verification scams, technical support fraud, and brand impersonation attacks—were examined. Each sample was analyzed using industry-recognized tools such as **Google Admin Toolbox**, **MXToolbox**, and **VirusTotal** to evaluate sender information, email authentication records (SPF, DKIM, and DMARC where available), message headers, suspicious domains, and embedded links. The analysis also considered social engineering techniques employed by attackers, including urgency, fear-based messaging, generic greetings, spoofed branding, and requests for immediate user action.

Based on the analysis, each email was classified according to its level of risk and the phishing indicators it exhibited. Common patterns identified across the samples included impersonation of trusted organizations, misleading subject lines, suspicious sender information, deceptive hyperlinks, and psychological manipulation designed to influence user behavior. These findings demonstrate how attackers combine technical deception with social engineering to increase the likelihood of successful phishing attacks.

In addition to technical analysis, this report provides user-focused awareness guidelines and best practices to help individuals recognize and respond appropriately to suspicious emails. The recommendations emphasize verifying sender identities, inspecting URLs before clicking, recognizing common phishing tactics, avoiding unsolicited requests for sensitive information, and reporting suspicious emails through appropriate organizational channels.

Overall, this project highlights the importance of combining technical email analysis with cybersecurity awareness to strengthen an organization's defense against phishing attacks. By understanding how phishing campaigns operate and educating users on common attack techniques, organizations can significantly reduce the risk of credential compromise, financial fraud, and unauthorized access to sensitive information.

2. Project Objective

The primary objective of this project is to analyze real-world phishing email samples to identify the technical and social engineering techniques commonly used by cybercriminals to deceive users. The project aims to develop a structured approach for detecting phishing emails by examining sender information, email headers, authentication mechanisms, embedded links, and suspicious content.

This project also seeks to enhance cybersecurity awareness by demonstrating how phishing attacks exploit trust, urgency, and brand impersonation to obtain sensitive information such as usernames, passwords, financial details, and one-time passwords (OTPs). By analyzing these attack patterns, the report provides practical guidance that enables users to recognize phishing attempts before becoming victims.

To achieve these objectives, the project focuses on the following:

- Collect and analyze multiple real-world phishing email samples representing different attack scenarios.
- Examine email headers, sender domains, authentication records (SPF, DKIM, and DMARC where available), and embedded URLs using industry-standard analysis tools.
- Identify technical indicators and social engineering techniques that distinguish phishing emails from legitimate communications.
- Classify each email based on its level of risk and provide evidence supporting the classification.
- Document the findings in a clear, structured, and professional security analysis report.
- Develop user awareness guidelines and security best practices to help individuals identify, avoid, and report phishing emails effectively.

Ultimately, the objective of this project is not only to detect phishing emails but also to promote cybersecurity awareness and strengthen an organization's first line of defense—the human user. By combining technical analysis with user education, this project demonstrates how effective phishing awareness can significantly reduce the risk of credential theft, financial fraud, and unauthorized access to organizational systems.

3. Scope of Analysis

This project focuses on the analysis of phishing emails from a cybersecurity awareness and email investigation perspective. The primary scope of the project is to examine real-world phishing email samples, identify indicators of compromise, evaluate the risk associated with each email, and provide practical recommendations to help users recognize and avoid phishing attacks.

The analysis includes the examination of **five phishing email samples**, each representing different phishing techniques such as credential theft, brand impersonation, account verification scams, and technical support fraud. Every email was analyzed using a structured methodology to assess its authenticity, identify suspicious characteristics, and determine the overall threat level.

The scope of this project includes the following activities:

- Collection and examination of five publicly available phishing email samples.
- Analysis of email components, including:
 - Sender information (From, Reply-To, Return-Path)
 - Subject line
 - Email content and formatting
 - Message headers
 - Message-ID
 - Email authentication records (SPF, DKIM, and DMARC where available)
- Identification of phishing indicators, including:
 - Brand impersonation
 - Generic greetings
 - Urgency and fear-based language
 - Suspicious hyperlinks
 - Social engineering techniques
 - Requests for sensitive information
- Risk classification of each email as **High, Medium, or Low** based on the observed phishing indicators.
- Investigation of embedded URLs and domains using security analysis tools where applicable.
- Documentation of findings in a structured security analysis report.
- Development of cybersecurity awareness recommendations and best practices to help users detect and report phishing attempts.

Out of Scope

To clearly define the boundaries of this project, the following activities were **not performed**:

- Interacting with or clicking on malicious links.
- Downloading or executing attachments from phishing emails.
- Exploiting vulnerabilities or conducting offensive security activities.
- Malware analysis or reverse engineering.
- Digital forensics on compromised systems.
- Live penetration testing of email infrastructure.

- Automated phishing detection using machine learning or artificial intelligence models.

This project is intended solely for **educational, defensive, and cybersecurity awareness purposes**. All phishing email samples were obtained from publicly available educational resources and were analyzed in a safe, controlled environment without interacting with malicious infrastructure.

4. Tools Used

The following tools and resources were utilized throughout this project to analyze phishing email samples, investigate email authenticity, inspect sender information, evaluate email authentication mechanisms, and assess the reputation of suspicious domains and URLs.

Tool / Resource	Purpose
Google Admin Toolbox – Message Header Analyzer	Used to analyze email headers, inspect the email delivery path, review SPF, DKIM, and DMARC authentication results, and identify routing information of the email.
MXToolbox – Email Header Analyzer	Used to perform detailed email header analysis, validate sender information, examine SMTP routing, and identify potential anomalies in email transmission.
VirusTotal	Used to check the reputation of URLs, domains, IP addresses, and files by comparing them against multiple security vendors and threat intelligence sources.
GitHub Public Phishing Email Repositories	Used as a source of publicly available phishing email samples for educational analysis and investigation.
Microsoft Word	Used to prepare, organize, and format the phishing analysis report in a professional documentation format.
Google Chrome	Used to safely access analysis tools and examine phishing email content without interacting with malicious links.

Phishing Analysis – Sample 1

● Amazon <olivia@amazonsupport.com>

November 24, 2017 at 12:39 PM



Alert

To:

Reply-To: Amazon <olivia@amazonsupport.com>



Password assistance

Someone tried to reset your password from **Dayton, Ohio**. If you have not requested this code

Please Call Us on [1-800-801-5811](tel:1-800-801-5811)

Please provide below mentioned code with your Email address to verify

161145

Amazon takes your account security very seriously. Amazon will never ask you to disclose or verify your Amazon password, credit card, or banking account number. If you receive a suspicious email with a link to update your account information, do not click on the link—instead, report the email to Amazon for investigation.

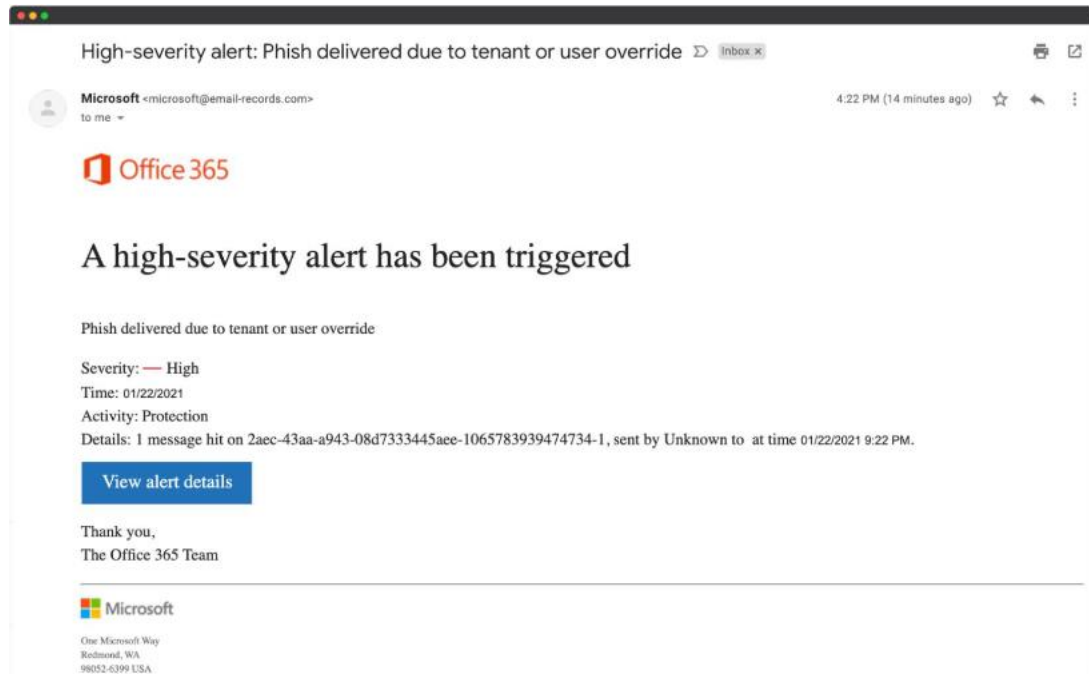
Indicator	Observation	Explanation
Brand Impersonation	The email uses Amazon's official logo and branding.	Attackers impersonate trusted organizations to make the email appear legitimate and increase the likelihood that recipients will trust the message.
Suspicious Sender Email	Sender address: olivia@amazonsupport.com	Although the domain appears related to Amazon, it is not the official Amazon domain (e.g., amazon.com). Attackers often register look-alike domains to deceive users.
Urgency / Fear Tactic	"Someone tried to reset your password..."	The email attempts to create panic by suggesting unauthorized access to the recipient's account, encouraging immediate action without verification.

Risk Classification

Risk Level: ● High Risk (Phishing)

This email exhibits multiple phishing characteristics, including impersonation of a trusted organization, fear-based messaging, a request for immediate action, and an attempt to redirect the victim to a fraudulent phone support service. These indicators strongly suggest that the email is designed to steal sensitive information through social engineering.

Phishing Analysis – Sample 2



Indicator	Observation	Explanation
Brand Impersonation	The email uses Microsoft's Office 365 logo, branding, and corporate design.	Cybercriminals frequently impersonate trusted organizations such as Microsoft to increase credibility and deceive recipients into trusting the email.
Suspicious Sender Email	Sender address: microsoft@email-records.com	The sender's domain is not an official Microsoft domain (e.g., microsoft.com or office.com). The use of a look-alike domain is a strong phishing indicator.
Urgent Subject Line	Subject: " High-severity alert: Phish delivered due to tenant or user override "	The subject is designed to create urgency and panic, encouraging the recipient to act immediately without verifying the email's authenticity.

Risk Classification

Risk Level: ● High Risk (Phishing)

This email exhibits multiple phishing characteristics, including impersonation of Microsoft Office 365, the use of a suspicious sender domain, fear-based security alerts, and a deceptive call-to-action button intended to lure victims into revealing their Microsoft account credentials. These indicators strongly suggest that the email is a credential phishing attempt.

Phishing Analysis – Sample 3

From: "MS-Support Centre" <outlook_2C5A4DD15A5A1106@outlook.com>
Sent: Tuesday, June 2, 2020 11:30:27 AM
Subject: Account unusual sign-in activity

Microsoft account

Suspicious Account Activity

Hello

This is to inform you that we have found suspicious activities with your account. Due to that, we have terminated your windows account.

Please find suspicious incident details:

Recent Incident Details: Eastern Belarus (IP Address : 10.97.87.25)
MAC Address 01:AD:99:00 & IP: 10.97.87.25

If you think this was a mistake and you wish to continue using this windows license key, Please contact consumer technical support at 1-800-341-8835.

PS NOTE: Please be at your computer while you call consumer technical support.

Windows Help
1-800-341-8835.

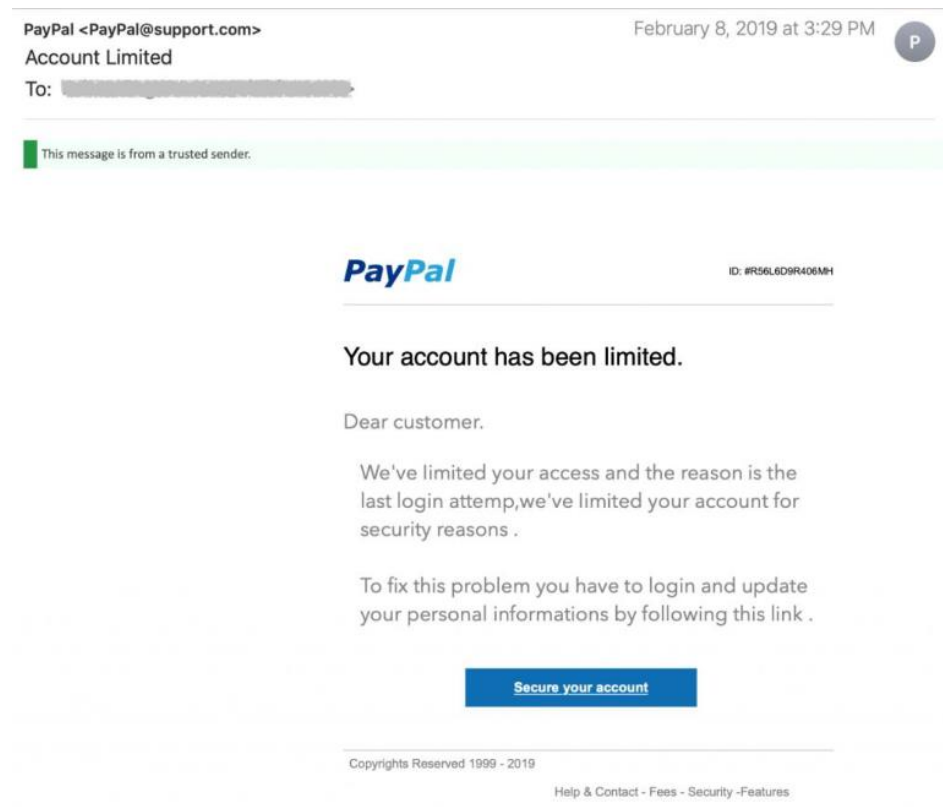
Indicator	Observation	Explanation
Brand Impersonation	The email uses Microsoft's name, branding, and "Microsoft Account" heading.	Attackers imitate trusted companies to gain the recipient's confidence and make the email appear legitimate.
Suspicious Sender Email	Sender address: 2C5A4DD15A5A1106@outlook.com	Official Microsoft security notifications are not sent from random Outlook email addresses. The sender address is inconsistent with legitimate Microsoft domains, making it highly suspicious.
Urgent Subject Line	Subject: " Account unusual sign-in activity "	The subject creates urgency by suggesting that unauthorized access has occurred, encouraging the recipient to react immediately.

Risk Classification

Risk Level: ● High Risk (Phishing)

This email displays several high-confidence phishing indicators, including Microsoft brand impersonation, a suspicious sender email address, fabricated security incident details, fear-based messaging, grammatical errors, and a fraudulent technical support phone number. The objective of the email is likely to convince the recipient to contact the attacker, enabling credential theft, financial fraud, or unauthorized remote access.

Phishing Analysis – Sample 4



Indicator	Observation	Explanation
Brand Impersonation	The email uses the official PayPal logo, branding, and color scheme.	Attackers imitate trusted companies to make the email appear authentic and increase the likelihood that recipients will trust the message.
Suspicious Sender Email	Sender address: PayPal@support.com	Official PayPal emails are sent from verified @paypal.com domains. The use of support.com instead of paypal.com is a strong indicator of impersonation.
Urgent Subject Line	Subject: " Account Limited "	The subject creates urgency by suggesting that the recipient's account has been restricted, encouraging immediate action

Phishing Analysis – Sample 5

MessageId	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Created at:	9/20/2023, 12:05:49 AM GMT+5:30 (Delivered after 56 sec)
From:	BANCO DO BRADESCO LIVELO<banco.bradesco@atendimento.com.br>
To:	phishing@pot
Subject:	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
SPF:	temperror with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	temperror Learn more

Header Analyzed

Email Subject: PRIME CLIENT - BRADESCO LIVELO: Your card has 92,990 LIVELO points expiring today!

← Analyze New Header

Copy/Paste Warning

Copying/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

- ✖ DMARC Compliant (No DMARC Record Found)
 - ✖ SPF Alignment
 - ✖ SPF Authenticated
 - ✖ DKIM Alignment
 - ✖ DKIM Authenticated

Indicator	Observation	Explanation
Brand Impersonation	The email claims to be from BANCO DO BRADESCO LIVELO .	Attackers impersonate well-known financial institutions to gain the recipient's trust and increase the likelihood of interaction.
Urgent Subject Line	Subject: " CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje! "	The subject creates urgency by claiming reward points are expiring today, encouraging recipients to act immediately without verifying the email's legitimacy.
Suspicious Sender Domain	Sender address: banco.bradesco@atendimento.com.br	Although the sender appears to reference Bradesco, the email originates from a different domain than the bank's official email infrastructure. Attackers frequently use look-alike or unrelated domains to impersonate legitimate organizations.
SPF Authentication Result	SPF: TempError	The sender's SPF authentication could not be successfully validated. While a TempError may occur due to DNS issues, it prevents verification that the sending server was authorized to send emails on behalf of the claimed domain.
DKIM Authentication Missing	DKIM: None	The email is not digitally signed using DKIM. Without a valid DKIM signature, the recipient cannot verify the integrity or authenticity of the message.
DMARC Authentication Result	DMARC: TempError	DMARC validation could not be completed successfully. Since DMARC relies on SPF and/or DKIM, the inability to authenticate increases the uncertainty surrounding the sender's legitimacy.
Financial Reward Lure	The email promises that 92,990 reward points are about to expire.	Attackers commonly exploit financial incentives and reward

7. Employee Awareness & Prevention Guidelines

Phishing attacks continue to be one of the most common methods used by cybercriminals to gain unauthorized access to sensitive information. Although organizations deploy advanced security technologies such as email filtering, antivirus software, and firewalls, human error remains a major factor in successful phishing attacks. Therefore, employee awareness and adherence to cybersecurity best practices are essential in reducing the risk of compromise.

Employees should follow the guidelines below when handling emails:

1. Verify the Sender's Email Address

Always inspect the sender's email address rather than relying solely on the display name. Cybercriminals often impersonate trusted organizations using look-alike or spoofed email addresses.

2. Be Cautious of Urgent or Threatening Messages

Emails claiming account suspension, password expiration, prize winnings, or financial penalties are designed to create panic. Employees should remain calm and verify such claims through official communication channels.

3. Inspect Hyperlinks Before Clicking

Before clicking any button or hyperlink, hover the mouse cursor over the link to verify the destination URL. Avoid clicking links that redirect to unfamiliar or misspelled domains.

4. Avoid Opening Unexpected Attachments

Do not open attachments received from unknown or unexpected senders, especially executable files, compressed archives, or Office documents requesting macros to be enabled.

5. Verify Requests for Sensitive Information

Legitimate organizations rarely request passwords, banking information, or multi-factor authentication (MFA) codes through email. Such requests should always be considered suspicious.

6. Check for Grammar and Formatting Errors

Many phishing emails contain spelling mistakes, poor grammar, inconsistent formatting, or unusual sentence structures that may indicate fraudulent communication.

7. Confirm Requests Through Official Channels

If an email requests financial transactions, password resets, or confidential information, verify the request by contacting the organization using official phone numbers or websites instead of responding directly to the email.

8. Report Suspicious Emails Immediately

Employees should report suspected phishing emails to the organization's IT or Security Operations Center (SOC) team. Prompt reporting allows security teams to investigate and prevent similar attacks from affecting other users.

9. Keep Systems Updated

Operating systems, browsers, email clients, and security software should be regularly updated to protect against known vulnerabilities exploited by phishing campaigns.

10. Participate in Security Awareness Training

Regular cybersecurity awareness programs help employees recognize emerging phishing techniques, social engineering tactics, and best practices for protecting organizational information.

Key Takeaways

- Always verify the sender before trusting an email.
- Never click suspicious links or download unexpected attachments.
- Be cautious of urgent requests involving money, passwords, or confidential information.
- Verify important requests through official communication channels.
- Report suspicious emails immediately to the organization's security team.
- Regular security awareness training significantly reduces the likelihood of successful phishing attacks.

Conclusion

This project provided a practical analysis of five phishing email samples to understand how cybercriminals use social engineering techniques to deceive users into revealing sensitive information. By examining each email individually, several phishing indicators such as brand impersonation, suspicious sender addresses, urgent language, misleading subject lines, fake security alerts, malicious links, and requests for confidential information were successfully identified.

The analysis demonstrated that phishing attacks often rely more on psychological manipulation than on technical exploitation. Attackers create a sense of urgency, fear, curiosity, or reward to encourage users to make quick decisions without carefully verifying the authenticity of the email. Even professionally designed emails that closely resemble legitimate communications can be identified by carefully examining the sender information, domain names, writing style, hyperlinks, and authentication mechanisms such as SPF, DKIM, and DMARC when available.

Throughout this project, structured analysis methodologies and publicly available security tools were used to evaluate the phishing samples and classify their risk levels. The findings emphasize that effective phishing detection requires a combination of technical analysis and user awareness. While email security technologies provide an important layer of defense, informed users remain one of the strongest protections against phishing attacks.

Finally, this project highlights the importance of cybersecurity awareness, continuous employee training, and safe email handling practices in reducing the success rate of phishing campaigns. As phishing techniques continue to evolve and become increasingly sophisticated, organizations and individuals must remain vigilant, verify unexpected communications, and adopt security best practices to protect sensitive information and digital assets.

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